

Project Plan: Socio-Economic & Demographic Analysis of Indian Festivals

1. Problem Statement

India exhibits immense cultural and regional diversity, with hundreds of festivals celebrated across various states, communities, and calendar months. However, unstructured information often leads to biased representations, lack of comparative insights, and underappreciated economic and demographic patterns. This project aims to systematically collect authentic multi-faith festival data, build predictive machine learning models, perform rigorous statistical evaluations, and communicate empirical insights publicly through a data-driven visualization blog and an executive video presentation.

2. Comprehensive Task Breakdown

Task 1: Comprehensive Data Collection & Preprocessing

- **Objective:** Curate an authentic, multi-dimensional, secular dataset on festivals across India.
- **Scope & Dimensions:**
 - **Temporal & Geographic:** Month-wise distribution, state/UT breakdown, duration, and seasonality.
 - **Socio-Cultural Inclusivity:** Multi-faith representation encompassing Hindu, Muslim, Christian, Jain, Sikh, Buddhist, Parsi, and regional/tribal festivals.
 - **Economic & Footfall Metrics:** Consumer spending trends, market sector impact (retail, travel, jewelry, FMCG), and estimated crowd sizes/trends where available.
- **Strict Constraints:**
 - **Zero Synthetic Data:** All records must be backed by real, verified sources (government portals, census data, economic survey reports, verified news archives).
 - **Source Grounding:** Maintain exact URL citations and attribution metadata for every record to ensure factual accuracy and avoid misinformation.
- **Key Deliverables:**
 - Raw dataset (CSV / JSON) with source attribution columns.
 - Cleaned and preprocessed tabular dataset ready for statistical analysis and modeling.
 - Data dictionary defining all schema attributes and collection sources.

Task 2: Machine Learning Model Development & Application

- **Objective:** Implement and evaluate machine learning models to identify patterns, classify categories, or predict economic/crowd trends from festival attributes.

- **Potential ML Implementations:**
 - **Regression/Forecasting:** Estimating economic expenditure or footfall trends based on duration, region, and historical indicators.
 - **Classification/Clustering:** Multi-label classification of festival typologies or clustering regional celebration patterns.
- **Key Deliverables:**
 - Documented ML pipeline code (data preprocessing, feature engineering, training, validation).
 - Model evaluation metrics (RMSE, R^2 , Precision, Recall, or Silhouette Score, depending on model type).
 - Inferences and actionable takeaways derived from model predictions.

Task 3: Statistical Valuation & Exploratory Data Analysis (EDA)

- **Objective:** Perform robust statistical testing and valuation on the collected festival data independent of ML inferences.
- **Key Analysis Areas:**
 - **Descriptive Statistics:** Measures of central tendency, dispersion, and distribution across spending, frequency, and duration.
 - **Hypothesis Testing & Correlation Analysis:** Evaluating relationships between festival timing, regional economic spikes, and footfall densities.
 - **Comparative Analysis:** Objective comparisons across states, seasons, and global analogues without subjective bias.
- **Key Deliverables:**
 - Statistical summary report (distribution plots, correlation matrices, significance test results).
 - Quantitative validation tables supporting the findings.

Task 4: Public Data Visualization Blog & Community Engagement

- **Objective:** Author and publish a public-facing, visually rich blog post presenting the data visualizations and socio-cultural insights.
- **Content & Visualization Requirements:**
 - **Charts & Visuals:** State-wise heatmaps, month-wise festival density timelines, demographic breakdowns, and economic generation charts.
 - **Tone & Framing:** Secular, balanced, and evidence-grounded; zero bias toward or against any specific community or religion.
 - **Mandatory Attribution:** Visible citations and footnotes for all referenced figures.
- **Platform & Outreach Requirements:**
 - **Publishing Platform:** LinkedIn Article, Medium, or a public blogging portal.
 - **External Engagement Target:** Minimum **40 external reactions/comments** from genuine, external professionals and readers (strictly excluding peers from your own institution).
 - **Interactive Engagement:** Active author participation responding to questions and discussions in the comment section.
- **Key Deliverables:**
 - Live URL of the published blog post.
 - High-resolution visualization graphics and charts.

- Engagement audit report (screenshots and metrics proving 40+ external interactions and active comment responses).

Task 5: Final Report Video & Synthesis

- **Objective:** Produce a structured summary video presenting the entire project pipeline, statistical findings, model evaluations, and blog impact.
- **Structure & Coverage:**
 1. Project motivation, dataset curation methodology, and source integrity.
 2. Statistical evaluation and exploratory data findings.
 3. Machine learning architecture, performance, and key takeaways.
 4. Overview of the published blog, visualizations, and external public response.
- **Key Deliverables:**
 - Final high-definition report video (MP4 file or hosted video link).
 - Complete project portfolio repository containing code, datasets, visual assets, and engagement logs.

3. Master Deliverables Matrix

Task	Primary Deliverable	Acceptance Criteria
Task 1: Data Collection	Validated Dataset & Data Dictionary	Multi-faith coverage, 100% verified sources, zero synthetic records.
Task 2: ML Pipeline	Model Code, Weights & Evaluation Report	Documented training workflow, quantitative validation metrics.
Task 3: Statistics	Statistical Valuation & EDA Report	Hypothesis test results, descriptive statistics, correlation matrices.
Task 4: Visualization Blog	Published Article & Engagement Proof	Public URL, interactive visualizations, \ge 40 external reactions/comments.
Task 5: Final Presentation	Synthesis Report Video	Structured presentation covering data, stats, ML, and blog reception.